This article has been accepted for publication in IEEE Transactions on Cognitive Communications and Networking. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2025.3565589

An Open-Set Few-Shot Class Incremental Learning Framework for Specific Emitter Identification

Tao Zhang, Xiaoyu Shen, Zhihui Shang, Guoru Ding, Hao Wu and Xiaoqiang Qiao

***Abstract*—Specific Emitter Identification (SEI) technology enhances the physical-layer security of the Internet of Things (IoT). However, traditional SEI systems primarily focus on identification tasks within static closed-set environments, limiting their applicability in dynamic open-set scenarios. To address the challenges of Few-Shot Incremental Open-Set Recognition (FIOSR), which involves the expansion of specific emitter categories and Open-Set Recognition (OSR), we propose an Open-Set Few-Shot Class Incremental Learning (OFSCIL) framework. Our framework decouples the feature extractor from the classifier, and optimizes the extractor using an innovative integrated loss function and class augmentation. Moreover, we design a sustainably evolving prototype classifier with a prototype calibration method to alleviate the catastrophic forgetting and overfitting in the Few-Shot Class Incremental Learning (FSCIL) phase. For recognizing unknown categories, we design an open-set classifier based on prototype distance. Additionally, a semi-supervised clustering algorithm using cosine distance is introduced to further distinguish unknown signals. Experimental results on the benchmark dataset show that OFSCIL is well adapted to OSR, FSCIL and FIOSR tasks, highlighting its capability to balance computational efficiency and accuracy.**

***Index Terms*—Specific emitter identification, few-shot incremental open-set recognition, class incremental learning, open-set recognition, radio frequency fingerprints**

I. Introduction

WITH the rapid advancement of Internet of Things (IoT) technology, the proliferation of wireless communication devices has surged markedly, posing substantial challenges to the security of both IoT systems and the physical layer of communication networks. Specific Emitter Identification (SEI) technology addresses these challenges by processing collected signals and extracting unique Radio Frequency Fingerprints (RFF), enabling direct identification of transmitters at the physical layer [1], [2].

In recent years, deep learning techniques have been able to achieve high-precision, end-to-end recognition due to their advantages in automatic feature extraction, processing complex signals, and exhibiting strong robustness [3], [4], which have attracted great attention. However, most SEI systems based on DL methods have been tailored for classification tasks within static, closed-set scenarios and are ineffective in dynamic open-set environments. To ensure that SEI systems operate effectively in open and dynamic environments, the closed-world assumption must be discarded. Recent research highlights the capabilities required for systems operating in open-world settings [5], which can be broadly categorized into two groups: (i) the ability to recognize whether an entity belongs to an unknown category (i.e., Open-Set Recognition (OSR)), and (ii) the ability to learn new categories to expand the system's knowledge base (i.e., Class Incremental Learning (CIL)). Given the complexity and variability inherent in electronic surveillance environments, practical application scenarios are inherently dynamic and open [6]. A long-term deployed SEI system in such an environment will inevitably encounter unknown categories and new classification demands. Therefore, SEI systems must be equipped with these capabilities, which undoubtedly pose significant challenges. Several scholars have already conducted extensive research on this topic. Zhang et al. [7] proposed a lightweight contrastive learning framework designed for radio frequency fingerprinting, with the objective of tackling incremental OSR task. Similarly, Liu et al. [8] presented an incremental learning and OSR framework for remote sensing image scene classification, employing a controlled convex hull-based example selection technique to mitigate catastrophic forgetting during incremental learning.

However, in most existing studies [7]–[9], while these systems are capable of effectively learning from new class data, they typically require a sufficient number of new class samples. In real-world scenarios, new categories frequently emerge, yet their sample sizes are often limited [10]. A system must be able to quickly update and adapt with only a few incremental class samples. Few-Shot Class Incremental Learning (FSCIL) [11] is used to cope with this problem. By rapidly learning and recognizing new classes while retaining previously acquired knowledge, FSCIL enables systems to better adapt to the dynamic changes of open environments. Compared to traditional incremental learning, this is clearly a more practical paradigm. For SEI systems, the newly introduced categories often have insufficient samples, making it a more challenging task to effectively learn the characteristics of new classes without

This work was supported in part by the Project of Natural Science Foundation of China (No.62371463, No.62171462, No.62231027, No.62425118), and the Key Technologies R&D Program of Jiangsu (Prospective and Key Technologies for Industry) under Grants BE2023022 and BE2023022-4.(Corresponding author: Xiaoyu Shen; Hao Wu)

Tao Zhang, Hao Wu, Xiaoqiang Qiao are with the Sixty-third Research Institute of National University of Defense Technology, Nanjing 210007, China; (e-mail:ztcool@126.com; whao1983@126.com; qxq0527@163.com).

Xiaoyu Shen is with the Sixty-third Research Institute of National University of Defense Technology, Nanjing 210007, China; and School of Electronic and Information Engineering, Nanjing University of Information Science and Technology, Nanjing 210044, China; (e-mail:18115113151@163.com).

ZhiHui Shang is with the Sixty-third Research Institute of National University of Defense Technology, Nanjing 210007, China; and the College of Medical Information Engineering, Zunyi Medical University, Zunyi 563000, China; (e-mail:szhldx@163.com).

Guoru Ding is with the College of Communications Engineering, Army Engineering University of PLA, Nanjing 210007, China (e-mail: guoru_ding@yeah.net).

experiencing forgetting under these conditions. Therefore, SEI systems deployed in dynamic open environments should have both FSCIL and OSR capabilities.

To counter the above challenges, this paper proposes a framework called Open-Set Few-Shot Class Incremental Learning (OFSCIL). OFSCIL employs a decoupled approach between the feature extractor and the classifier to mitigate catastrophic forgetting, utilizing a carefully designed unbiased prototype classifier as an alternative to prevent overfitting on small samples. During training, we propose a novel integrated loss function and a class augmentation strategy to ensure robust feature representation. In the FSCIL phase, the unbiased prototype classifier handles the continuously increasing number of signal categories, and a prototype calibration method is proposed to accurately characterize the prototypes of new signal categories. Moreover, an open-set classifier based on prototype distance identifies unknown categories that appear in the testing phase. Furthermore, we introduce a semi-supervised k-means++ (ss-k-means++) algorithm based on cosine distance to further distinguish unknown signals.

Ultimately, the proposed OFSCIL framework enables SEI systems to perform class incremental learning under few-shot conditions while also recognizing unknown classes, making it suitable for Few-Shot Incremental Open-Set Recognition (FIOSR) task. The main contributions of this paper are summarized as follows:

- Unlike existing research, this paper proposes a framework for SEI named OFSCIL, taking into account the FIOSR scenario that integrates OSR and FSCIL. The framework decouples feature extractor and classifier, providing a low-complexity solution for SEI systems.
- We propose a novel integrated loss function and a signal class augmentation strategy to optimize feature representation. In addition, we introduce a prototype calibration method to accurately represent the feature prototype.
- We design a prototype distance-based open-set classifier to identify unknown signals, and a semi-supervised k-means++ algorithm based on cosine distance to further differentiate them.

The remainder of this paper is organized as follows: Section II reviews related works on SEI. Section III defines the problem statement we aim to solve. Section IV presents the proposed OFSCIL framework in detail. Section V provides the evaluation experiments and analysis of the results. Section VI concludes the paper.

II. RELATED WORK

A. Open-set recognition

In open-world scenarios, the training dataset often does not contain all possible emitter categories, and the system may encounter unknown categories during testing. This requires the model to not only recognize known categories but also reject unknown, unseen, or novel categories [12]. The three main strategies for OSR right now are the one-vs-all approach, threshold techniques, and estimating the probability of unknowns [13]. Several studies have explored OSR tasks within the context of SEI. Huang et al. [14] utilized a denoising autoencoder, supplemented by center loss, to achieve open-set recognition of specific transmitters. Guo et al. proposed a triple anomaly detection mechanism to identify anomalous transmitters in [15]. Zhang et al. [16] calculated optimal open-set decision boundaries by combining a fixed softmax threshold [17] with weibull calibration. Zhang et al. [18] then introduced the extreme value theory. The effective operation of the SEI system in an open-world setting hinges on its ability for open-set recognition.

B. Class incremental learning

In practical applications, the types of specific emitters may continuously increase over time. Traditional DL methods often require retraining on the entire dataset [19], [20], while Class Incremental Learning (CIL) enables the model to progressively assimilate new classes while preserving knowledge of previously learned classes, thus saving time and computational resources [21]. In recent years, researchers have conducted several studies on CIL for SEI. Xu et al. [22] proposed a tri-way incremental learning algorithm to accommodate the growth of transmitter features, types, and samples. Liu et al. [23] updated the model by integrating knowledge distillation [24] with Elastic Weight Consolidation (EWC) [25]. However, catastrophic forgetting remains a challenge [26]. To address this, Liu et al. [27] maintained the memory of old classes by merging new samples with replayed samples of old classes, although the storage of replayed samples requires additional memory space. In [28], Zhou et al. introduced a distillation incremental learning method that updated the entire model based on a continuous data stream. Shi et al. [29] proposed a method for managing variable-length signal inputs and utilized incremental learning to achieve real-time data stream processing. Li et al. [30] proposed a CIL method that combines knowledge distillation with weight alignment, along with a prototype enhancement strategy. The key to effective operation of SEI systems in the open world is the ability to recognize open sets. Although some progress has been made in CIL, research on Open-Set CIL and FSCIL in the context of SEI remains limited [7], [31]. Zhao et al. [32] proposed a framework for anomaly device identification and tracking. This framework is capable of detecting anomalous devices and incrementally learning the characteristics of anomalous emitters, thereby updating the classifier accordingly. Li et al. [33] implemented few-shot class-incremental transmitter identification by introducing prototype learning, self-supervised contrastive learning, and a three-stage incremental learning approach. In [34], Li et al. addressed the FIOSR problem using meta-learning, but their approach requires a large number of initial classes (over 200 classes) to construct the meta-learner. SEI should be able to operate effectively in more complex, open and dynamic environments. Therefore, FIOSR combining FSCIL and OSR is the key to cope with this problem. The research in this paper focuses on FIOSR scenarios.

III. PROBLEM STATEMENT

As shown in Fig.1, this paper presents a more robust paradigm for SEI systems operating in open-world environments, requiring them to perform OSR while also adapting to the FSCIL task. OSR becomes a basic requirement of SEI for classifying known emitters and detecting unknown emitters in real scenarios. At the same time, the SEI should be scalable in dynamic environments for tasks where new emitters can be added with only a few number of samples. Fig.2 shows the three problems: OSR, FSCIL, and FIOSR.

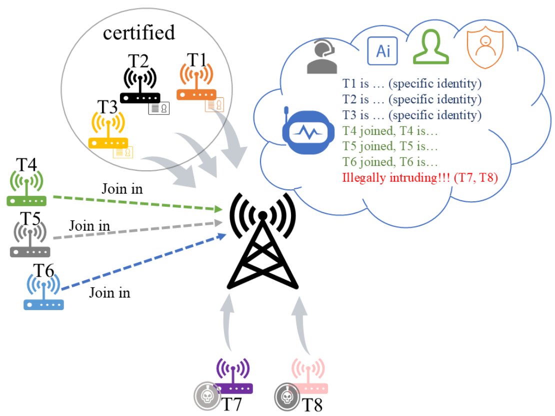


Fig. 1: A SEI system operating in an dynamic open-world

1) OSR: In the OSR task, unknown classes only appear during the testing phase. Given a training dataset $D_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$, where x_i represents the input samples and y_i denotes the corresponding known class labels, $y_i \in C_{\text{known}}$, N represents the sample count in the training set. The test dataset $D_{\text{test}} = \{(x_j, y_j)\}_{j=1}^{M}$ $(M > N)$ contains samples with known class $y_j \in C_{\text{known}}$ and unknown classes $y_j \in C_{\text{unknown}}$, where $C_{\text{unknown}} \cap C_{\text{known}} = \emptyset$ and M is the sample count in the test set. The objective of the OSR task is for the model to not only accurately classify samples where $y_j \in C_{\text{known}}$, but also to recognize samples where $y_j \in C_{\text{unknown}}$ as belonging to unknown class.

2) FSCIL: FSCIL aims to progressively extend a model's classification capacity with a limited amount of new class data. In FSCIL, the model starts with an initial training phase that includes a base training dataset with a substantial quantity of samples and classes. Then, in the incremental learning phases, the model learns new classes sequentially, but the training samples available for each new category are quite limited. The setup is defined as follows:

Let the base training dataset be denoted as $\mathcal{D}_{\text{base}} = \{(\mathbf{x}_i, y_i) \mid i = 1, 2, \ldots, N_{\text{base}}\}$, where $\mathbf{x}_i$ represents the feature vector of the sample, and $y_i \in \mathcal{C}_{\text{base}}$ denotes the corresponding class label, with $\mathcal{C}_{\text{base}}$ being the set of base classes. The incremental training dataset for the t-th phase is denoted as $\mathcal{D}_{\text{inc}}^{t} = \{(\mathbf{x}_j, y_j) \mid j = 1, 2, \ldots, N_{\text{inc}}^{t}\}$, where $\mathbf{x}_j$ represents the feature vector of the sample, and $y_j \in \mathcal{C}_{\text{inc}}^{t}$ denotes the class label of the t-th incremental phase, with $\mathcal{C}_{\text{inc}}^{t}$ representing the set of new classes in the t-th phase. During each incremental phase, the model must learn the new classes based on $\mathcal{D}_{\text{inc}}^{t}$ while maintaining good recognition performance on the old classes, defined as $\mathcal{C}_{\text{base}} \cup \bigcup_{i=1}^{t-1} \mathcal{C}_{\text{inc}}^{i}$. Specifically, the model's objective is to minimize the following loss function:

$$\mathcal{L} = \sum_{(\mathbf{x},y)\in\mathcal{D}_{\text{base}}\cup\bigcup_{i=1}^{t}\mathcal{D}_{\text{inc}}^{i}} \ell(f(\mathbf{x};\theta), y), \tag{1}$$

where $\ell(\cdot)$ denotes the loss function, and $f(\cdot;\theta)$ represents the model with its parameters θ.

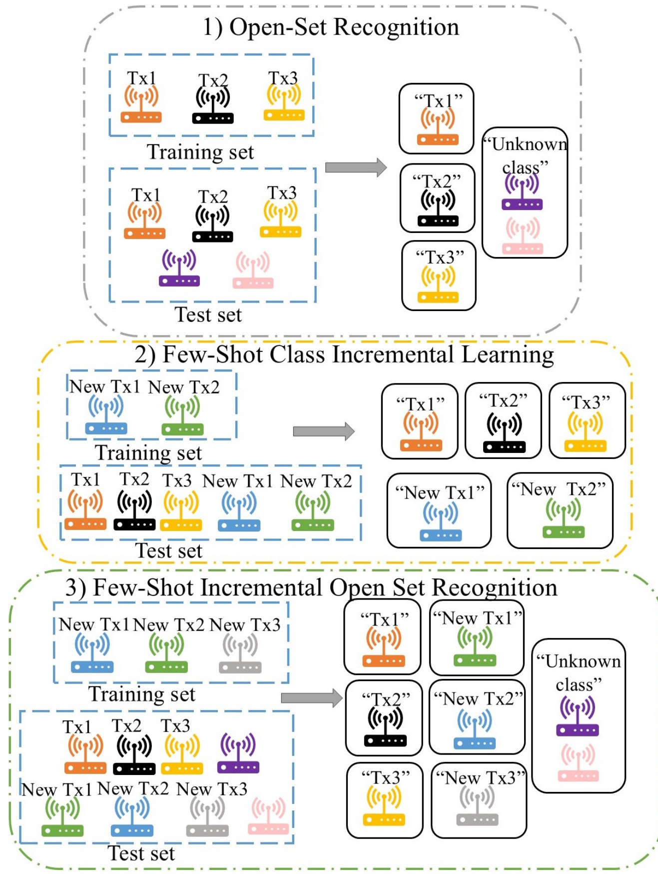


Fig. 2: Descriptions of 1) OSR, 2) FSCIL and 3) FIOSR.

Compared to traditional CIL task, FSCIL faces challenges beyond catastrophic forgetting. Given the limited number of new class samples, the model must extract effective information from few samples to learn new classes while also taking steps to prevent overfitting. The scarcity of new class samples makes the FSCIL task more challenging than CIL, as it must address both catastrophic forgetting and overfitting simultaneously.

3) FIOSR: Building on the previous analysis, the FIOSR task combines the challenges of OSR and FSCIL. Specifically, FIOSR requires a model to not only learn from limited samples of new categories but also to possess the ability to recognize unseen categories during the testing phase. Let the base training set be denoted as $\mathcal{D}_{\text{base}} = \{(\mathbf{x}_i, y_i) \mid i = 1, 2, \ldots, N_{\text{base}}\}$, and the incremental training dataset for the t-th phase as $\mathcal{D}_{\text{inc}}^{t} = \{(\mathbf{x}_j, y_j) \mid j = 1, 2, \ldots, N_{\text{inc}}^{t}\}$. The test dataset includes unknown classes from $\mathcal{C}_{\text{unknown}}$. The model is required not only to correctly classify samples from $\mathcal{C}_{\text{base}} \cup \bigcup_{i=1}^{t} \mathcal{C}_{\text{inc}}^{i}$, but also to recognize samples from $\mathcal{C}_{\text{unknown}}$. At the same time, it must prevent overfitting and catastrophic forgetting, ensuring the preservation of recognition ability for old classes as new ones are introduced.

The complexity of the FIOSR task stems from its concurrent involvement in the challenges of FSCIL and OSR. In FSCIL tasks, the model encounters the challenge of incremental

This article has been accepted for publication in IEEE Transactions on Cognitive Communications and Networking. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2025.3565589

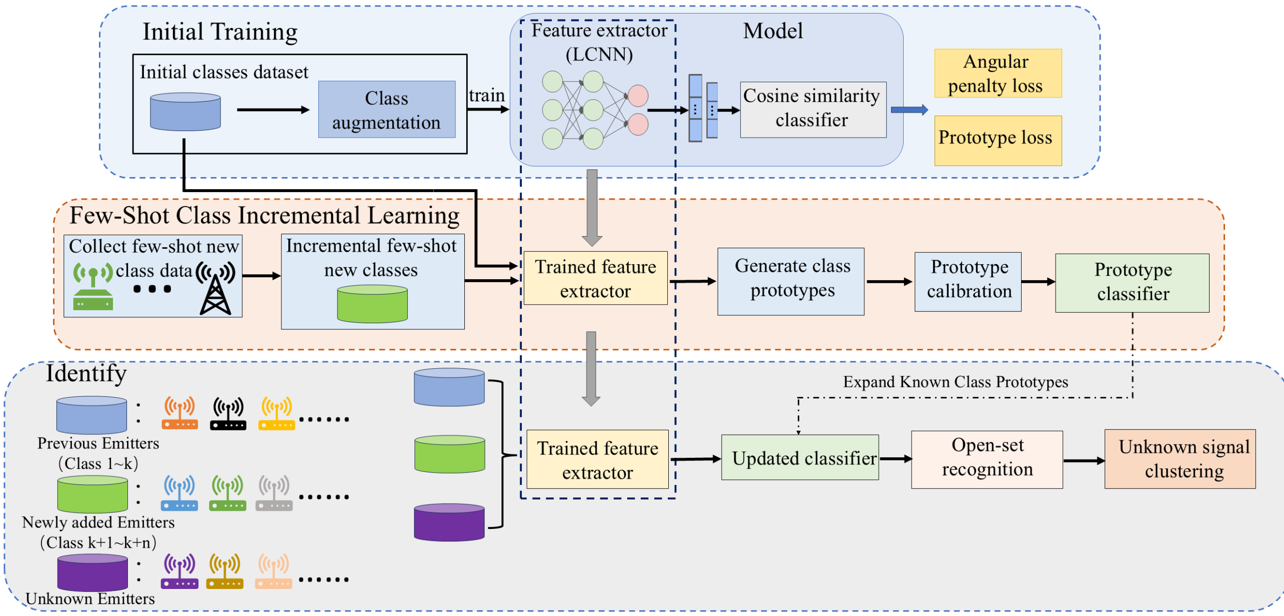


Fig. 3: Structure of OFSCIL framework

learning, while in OSR tasks, the model must differentiate between known and unknown categories. FIOSR integrates both of these challenges, requiring the model to handle limited samples during the incremental learning phase, while also identifying unknown categories in the testing phase. This combination makes FIOSR a more complex task than FSCIL or OSR in isolation.

IV. OUR PROPOSED METHOD

OFSCIL proposed in this paper is to help SEI systems cope with dynamic open-set environments. This paper adopts a strategy that decouples the feature extractor from the classifier, overcoming the limitations of the output dimensions in traditional DL networks by allowing flexible adjustment based on classification needs [35]. The frozen feature extractor prevents the loss of historical knowledge, while the designed prototype classifier reduces the risk of overfitting [36]. The proposed OFSCIL framework is shown in Fig.3.

A. Framework overview

OFSCIL consists of three phases: initial training, FSCIL, and identify. The initial training phase focuses on training a robust feature extractor to capture effective RFF features for each specific emitter, thereby generating accurate feature prototypes for each initial emitter class. We propose a methodology for training a feature extractor that incorporates an enhanced angular penalty loss in conjunction with a prototype loss. The combination of these losses improves the model's feature representation capabilities by enhancing inter-class separability and intra-class compactness. Class augmentation is proposed to enhance the model's generalization capability by generating new class data, mitigating the impact of class imbalance and optimizing feature representation.

In the FSCIL phase, the pretrained feature extractor is used to extract features of newly added emitter types. An unbiased nearest class mean classifier, also known as a prototype classifier, generates representative feature prototypes for each class. Considering the difficulty of generating accurate feature prototypes for new classes with few samples, a prototype calibration method is proposed. The prototype classifier then identifies the old and new categories. Each incremental learning step integrates new class prototypes into the known class prototypes, thereby achieving the continuous evolution of the classifier's knowledge. It is worth noting that if no new specific emitters are introduced, the FSCIL phase can be skipped.

In the identify phase, the open-set classifier inherited from the prototype classifier plays a crucial role in identifying known emitters and detecting unknown ones. OSR is effectively achieved by computing the cosine distance between the feature prototypes of known classes and the test signals. The ss-k-means++ algorithm based on cosine distance, further differentiates the detected unknown signals.

B. Initial training

The initial training phase aims to train the model using the initial class to obtain a robust feature extractor that can generate robust feature representations. In this paper, we design a combined loss function and propose a class enhancement strategy for training the feature extractor.

1) Loss function: This paper proposes an improved method that combines angular penalty loss with prototype loss. In the initial training stage, the compact feature distribution helps reduce interference from old-class features during subsequent incremental learning, thereby mitigating the risk of catastrophic forgetting. To further enhance the effectiveness of incremental learning, the use of a combined loss function optimizes both intra-class compactness and inter-class separability. More compact intra-class clustering and broader inter-class separation allow the model to allocate sufficient space in the feature space for new classes, thus reducing the confusion between old and new classes.

By introducing angular distance constraints in the feature space, angular penalty loss [37] significantly enhances the intra-class compactness and inter-class separability, thereby effectively improving the discriminative power of features. By imposing stricter angular constraints on intra-class samples, angular penalty loss helps reduce the spread of intra-class features while expanding the angular differences between categories. This allows the model to be more robust in handling previously unseen categories and better adapt to the introduction of new categories in incremental scenarios. In this paper, we design an enhanced angular penalty loss based on ArcFace. ArcFace [38] enhances the model's discriminative ability by introducing an angular penalty term at the classification boundary. Unlike the original ArcFace loss, which uses a fixed angular margin between feature vectors and class centers to improve classification performance, our method introduces an adaptive margin. This margin dynamically adjusts based on the characteristics of the samples, making it more flexible and better suited to different sample distributions. The enhanced angular penalty loss function is as follows:

$$L_{lap} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{e^{s\cos(\theta_{y_i}+\text{adm})}}{e^{s\cos(\theta_{y_i}+\text{adm})}+\sum_{j\neq y_i}e^{s\cos(\theta_j)}}, \quad (2)$$

where s is a scaling factor that amplifies the cosine similarity to improve the model's training efficiency and discriminative ability. θ_{y_i} represents the angle between the input feature vector of the i-th sample and its corresponding true class weight vector, which is computed as follows:

$$\theta_{y_i} = \arccos(\cos(\theta_{y_i})), \quad (3)$$

where $\cos(\theta_{y_i})$ represents the corresponding cosine similarity. Cosine similarity is a similarity measure based on the angle between two vectors. Let the feature vector of the sample be $\mathbf{f}_i$ and the weight vector be $\mathbf{W}_{y_i}$, then the cosine similarity is given by:

$$\cos(\theta_{y_i}) = \frac{\mathbf{f}_i \cdot \mathbf{W}_{y_i}}{\|\mathbf{f}_i\|\|\mathbf{W}_{y_i}\|}. \quad (4)$$

The term adm in Eq.2 represents the adaptive margin, which is calculated as follows:

$$\text{adm} = m\left(1+\lambda\cos(\theta)\right), \quad (5)$$

where the adaptive margin dynamically adjusts based on cosine similarity in addition to the fixed margin m, making the loss function more flexible to variations in the feature space. This enhances the model's training stability.

Prototype loss [39] optimizes the model by evaluating the similarity between sample features and their respective class prototypes, ensuring that the characteristics of samples within the same category exhibit greater similarity whereas those from different categories become more distinct, thereby improving classification performance. The prototype loss introduced in this paper achieves this by reducing the proximity between the sample features and their corresponding class prototypes, while maximizing the distance separating the sample features and the prototypes of other classes [40], as shown in the following:

$$L_{pro} = \frac{1}{N}\sum_{i=1}^{N}\max\left(0, \max_{j\neq y_i}(\mathbf{f}_i\cdot\mathbf{p}_j) - (\mathbf{f}_i\cdot\mathbf{p}_{y_i}) + b\right), \quad (6)$$

where $\mathbf{f}_i$ denotes the normalized feature vector of the i-th sample, and $\mathbf{p}_j$ signifies the normalized prototype vector of the j-th class. y_i is the true class label of the i-th sample. The term $\max_{j\neq y_i}(\mathbf{f}_i\cdot\mathbf{p}_j)$ represents the maximum cosine similarity between the i-th sample and all non-true class prototype vectors. $\mathbf{f}_i\cdot\mathbf{p}_{y_i}$ denotes the cosine similarity between the i-th sample and its true class prototype vector. The function $\max(0,\cdot)$ ensures that the loss value is non-negative. b is a hyperparameter that controls the minimum margin separating the true class from the closest incorrect class.

Feature prototypes typically refer to the center or representative vector of a class [41]. After each training iteration, the prototype vectors are updated to better represent the central features of each class. During the prototype update process, the feature vectors are first normalized. Subsequently, the prototype vector for each class is updated to equal the mean of the feature vectors of samples within that class:

$$\mathbf{p}_j = \frac{1}{|C_j|}\sum_{i\in C_j}\mathbf{f}_i, \quad (7)$$

where C_j denotes the group of samples associated with the class j. Finally, the total loss for training the feature extractor is given by the following formula:

$$\mathbf{L} = \lambda_1 L_{lap} + \lambda_2 L_{pro}. \quad (8)$$

2) Class augmentation: Class augmentation denotes the process of enhancing the model's generalization capability by expanding the number of classes within the training dataset. Through class augmentation, the feature extractor can generate more compact intra-class clusters and greater inter-class separations, leading to better feature distribution in the latent feature space, leaving more room for new incremental classes. Additionally, generating new class data enables the feature extractor to learn more diverse and transferable features. [42] and [43] propose randomly weighted combinations of different categories to synthesize auxiliary new category data. Furthermore, [30] proposes a method to enhance old class prototypes using gaussian white noise. Inspired by these methods, the proposed class augmentation method in this paper is shown as follows:

$$\mathbf{x}_{\text{aug}} = \alpha(\mathbf{x}_a + \mathbf{n}) + (1-\alpha)\mathbf{x}_b, \quad (9)$$

where $\mathbf{x}_a$ and $\mathbf{x}_b$ are two original signals, $\mathbf{n}$ is added gaussian noise that follows a normal distribution $\mathcal{N}(0,\sigma^2)$, and α is the mixing ratio coefficient.

3) Feature extractor: A lightweight one-dimensional convolutional neural network is designed as the feature extractor in our paper, and its structure is shown in Fig.4 and Table I. First, a one-dimensional convolutional layer is used to extract features related to IQ signals, followed by additional one-dimensional convolutional layers to capture more detailed features. Since dimensionality reduction is achieved in the initial stage, the computational load of feature extraction in subsequent layers is significantly reduced.

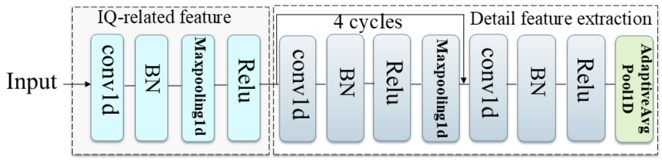


Fig. 4: Structure of Feature extractor

TABLE I: The corresponding hyper-parameters setting

Layer	Description	Hyper-parameters
IQ-related feature	Conv1D+BN	Filter=38, Kernel=2, Stride=1
	MaxPooling1D+ReLU	Kernel=2, Stride=2
Detail feature extraction	Conv1D+BN+ReLU	Filter=38, Kernel=3, Stride=1
	MaxPooling1D	Kernel=2, Stride=2
	Conv1D+BN+ReLU	Filter=38, Kernel=3, Stride=1
	AdaptiveAvgPool1D	——

Furthermore, due to the use of angular penalty and prototype loss instead of cross-entropy loss for model updating, we adopt a cosine similarity classifier for classification instead of the traditional fully connected layer, i.e., angular embedding is used instead of the traditional linear transformation. The traditional fully connected layer combined with cross-entropy loss focuses on probability maximisation but lacks explicit constraints on the geometric distribution of features in the embedding space. In addition, the weight parameters of the traditional fully connected layer may be biased (in favour of the base class) due to the fact that the base class has a much larger amount of data than the incremental class. Cosine similarity classifier computes the cosine similarity between feature vectors and normalises the weights and features, which can eliminate the effect of the size of the input features on the classification results, thus focusing only on the angular relationship between them and weakening the interference of the absolute values of the parameters. The cosine similarity classifier is only used to update the feature extractor, the formula is as follows:

$$P(c=i|x)=\frac{e^{\frac{\phi(x)^T w_i}{\|\phi(x)\|\|w_i\|}}}{\sum_{j=1}^{C} e^{\frac{\phi(x)^T w_j}{\|\phi(x)\|\|w_j\|}}}, \tag{10}$$

where $\phi(x)$ represents the feature vector of input x, w_i represents the weight vector related to class i, C represents the total number of classes.

C. Few-Shot class incremental learning

In the initial training phase, a robust feature extractor ϕ_θ is trained to effectively extract feature representations of the categories. In the FSCIL phase, the representation of each newly added class no longer relies on retraining the entire model but is instead achieved through feature prototypes which refers to the central representation of classes. This approach only requires calculating the prototype of the new class within the existing feature space, without the need for repeatedly updating the entire model, which significantly reduces computational complexity. Moreover, using a fixed feature extractor and feature prototype strategy competently alleviates the problem of catastrophic forgetting. Traditional approaches often necessitate updating the entire model to incorporate new classes, which can result in the loss of previously learned knowledge. In contrast, the feature prototype method mitigates catastrophic forgetting by preserving the feature representations of old classes; it achieves this by keeping the feature extractor fixed and introducing new prototypes only for the newly added classes. In the FSCIL scenarios, the sample size for new classes is typically very limited, and directly training the model on these few samples may lead to severe overfitting. However, the use of the feature prototype method can alleviate this overfitting issue to some extent. This is because the feature prototypes for the new classes are computed within an already stabilized feature space, without relying on retraining.

In the FSCIL phase, this paper designs a prototype classifier supplemented by prototype calibration method, for dealing with incremental new class.

1) Prototype generation: In this paper, we use the unbiased nearest class mean to represent the class feature prototype. Specifically, for the incremental class, all training samples of class c are used to compute the prototype [41] according to Eq.11.

$$\phi(c)=\frac{1}{n}\sum_{i=1}^{n}\phi(c_i), \tag{11}$$

whereas for the initial class, Firstly, the prototype $\phi(c)$ is computed using all the training samples. Then, for each sample feature $\phi(c_i)$, the cosine distance between it and the feature mean $\phi(c)$ is computed. The average of the top n samples with the smallest distances is selected as the final class prototype $\tilde{\phi}(c)$, n is the number of incremental class training samples. The final initial class prototype $\tilde{\phi}(c)$ is calculated as follows:

$$\tilde{\phi}(c)=\frac{1}{n}\sum_{j\in\mathcal{I}^*}\phi(c_j), \tag{12}$$

where $\mathcal{I}^*$ is the set of n sample features closest to the prototype, its formula is as follows:

$$\mathcal{I}^*=\underset{\mathcal{I}\subseteq\{1,2,\dots,N\},|\mathcal{I}|=n}{\arg\min}\sum_{i\in\mathcal{I}}\left(1-\frac{\phi(c_i)^\top\phi(c)}{\|\phi(c_i)\|\|\phi(c)\|}\right). \tag{13}$$

2) Prototype calibration: Due to the limited amount of data for newly added classes, the feature prototype for new classes can easily deviate from the true distribution of the class, meaning the calculated prototype may fail to accurately represent the core characteristics of the class. Moreover, the model is typically pre-trained on a substantial set of base classes, making the feature space for these base classes highly stable and possessing strong discriminative power. However, this strong discriminative power may also cause samples from new classes to be incorrectly attracted to similar base classes during classification, rather than being correctly identified as new classes. This phenomenon can further exacerbate the deviation of the new class prototypes. [36] found that prototypes of base classes trained on a large number of samples tend to be more stable and accurate, carrying rich feature information that helps adjust the deviation in new class prototypes. Even though the feature extractor is trained only on base classes, it is still capable of capturing feature similarities between base and new classes. This implies that certain base class prototypes may be very close to new incremental class prototypes in the feature space, and utilizing this similarity can help calibrate new incremental class prototypes, making them closer to the true distribution.

This paper introduces a prototype calibration method [36] to address this issue. By fusing the new class prototype with a weighted combination of similar base class prototypes, the distinguishability of the new class can be enhanced, resulting in a more accurate calibrated prototype. This is a two-stage calibration strategy. The first stage involves calibrating between the base classes and the new classes, with their cosine similarity given by:

$$\cos(\phi(c_b), \phi(c_n)) = \frac{\phi(c_b) \cdot \phi(c_n)}{\|\phi(c_b)\| \|\phi(c_n)\|}, \tag{14}$$

where $\phi(c_b)$ represents the prototype of a base class, and $\phi(c_n)$ represents the prototype of a newly added class. The cosine similarity between the new class c_n and the base class c_b is used to compute the weight $w_{b,n}$, which indicates the influence of the base class prototype in correcting the new class prototype. The weight is calculated as follows:

$$w_{b,n} = \frac{\exp(\tau \cdot \cos(\phi(c_b), \phi(c_n)))}{\sum_{j=1}^{B} \exp(\tau \cdot \cos(\phi(c_j), \phi(c_n)))}, \tag{15}$$

where the temperature parameter τ controls the sharpness of the weight distribution. The correction term Δc_b for the new class prototype based on the base class prototypes is computed according to the weights $w_{b,n}$, as shown:

$$\phi(\Delta c_b) = \sum_{b=1}^{B-1} w_{b,n} \phi(c_b). \tag{16}$$

The final calibrated prototype for the new class $\tilde{c}_n$ is given by Eq.17, where λ serves as a parameter that regulates the strength of calibration.

$$\phi(\tilde{c}_n) = (1-\lambda)\phi(c_n) + \lambda\phi(\Delta c_b). \tag{17}$$

To further enhance the distinguishability between new classes, a secondary calibration is performed among the newly added classes. The final calibration formula is as follows:

$$\phi(\bar{c}_n) = (1-\lambda)\phi(\tilde{c}_n) + \lambda \sum_{n'=1}^{N-1} w_{n',n} \phi(\tilde{c}_{n'}). \tag{18}$$

The final calibrated prototype incorporates both the connections linking the new classes with the base classes and the interconnections among the new classes, resulting in a more representative feature prototype for the incremental classes.

3) Prototype classifier: This paper decouples the feature extractor from the initial training model and designs a prototype classifier [44] as a substitute for the original classifier. For a test sample x, the cosine distance between its embedded feature $\phi(x)$ and each class prototype $\phi(c)$ is computed, and the class c with the shortest distance is selected as the prediction result. The final classifier is expressed as:

$$\hat{y} = \arg\min_{c \in C} \left(1 - \frac{\phi(x) \cdot \phi(c)}{\|\phi(x)\| \|\phi(c)\|}\right), \tag{19}$$

where C represents the set of all categories, including base classes c_b and incremental classes c_n. The designed prototype classifier is actually dynamic, each time a new class feature prototype is calculated, the new class prototype is included in the known class prototypes to continuously expand the knowledge of the classifier. This operation enables continuous evolution of the classifier.

D. Open-set recognition

In the identify phase, not only the initially trained categories and incremental new categories should be recognised, but also the unknown signals should be able to be rejected. In this paper, we design an open-set classifier for detecting unknown classes, which is inherited from the incremental prototype classifier. We also explore how to further distinguish the identified unknown signals.

1) Open-set classifier: For OSR, a threshold-based classifier is a popular approach for evaluating whether a sample corresponds to a known category [12]. Inspired by this, this paper designs a prototype-distance-based classifier. Specifically, the classifier determines whether a sample is part of a known category or an unknown category by computing the cosine distance linking the validation data with the prototypes of known classes, and making a decision based on a dynamic threshold. The cosine distance is computed as follows:

$$\text{cosine\_distance} = 1 - \frac{\phi(x) \cdot \phi(c)}{\|\phi(x)\| \|\phi(c)\|}, \tag{20}$$

where $\phi(x)$ represents the feature vector of the sample, and $\phi(c)$ represents the known class prototype. We determine the threshold by fitting a probability distribution.

For each sample in the training dataset, the cosine distance is computed between the sample's feature vector and each known class prototype. Then, the smallest cosine distance from all the category prototypes is chosen as the distance metric for that sample, because the minimum distance indicates the category to which the sample is most likely to belong. The mathematical expression is as follows:

$$\mathrm{d}_i = \min_c \left(\text{cosine\_distance}(\phi(x_i), \phi(c))\right), \tag{21}$$

where d_i is the minimum distance from sample i to the prototypes of all classes, where $\phi(x_i)$ is the normalized feature vector of sample i, and $\phi(c)$ represents the prototype of class c.

To differentiate between samples of known and unknown classes, we fit a normal distribution using the minimum cosine distances of the samples in the training dataset. These minimum distances are assumed to fit a normal distribution, denoted as:

$$\mathrm{ds} \sim \mathcal{N}(\mu, \sigma^2), \tag{22}$$

where ds is the set of the minimum cosine distances of all samples in the training data, μ and σ are the mean and standard deviation of the minimum distance. By fitting a normal distribution, we can dynamically determine the classification threshold based on the distribution of the training data.

Finally, the dynamic threshold is computed as:

$$\mathrm{dyt} = \mu + \sigma \cdot \Phi^{-1}(\alpha), \tag{23}$$

where Φ^{-1} is the quantile function of the normal distribution, μ and σ represent the mean and standard deviation of the minimum distances for the known classes, respectively. α represents the confidence level. The decision rule for the open-set classifier is as follows:

$$\text{prediction} = \begin{cases} \text{Known} & \text{if } \mathrm{d}_i < \mathrm{dyt} \times k \\ \text{Unknown} & \text{otherwise} \end{cases}, \tag{24}$$

where k is an adjustable scaling factor to control the stringency of the threshold. A larger k will make the classifier more lenient and may reduce the sensitivity to unknown classes, while a smaller k will make the classifier stricter and improve the recognition of unknown classes. When the test sample is identified as a known class, it is classified using the prototype classifier of Eq.19.

The detailed implementation process of OFSCIL, which is used for FIOSR, is shown in Algorithm 1.

2) Unknown signals cluster: The open-set classifier effectively identifies signals from known devices and labels unknown signals as the "unknown" class. However, simply categorizing all unknown signals into a single class fails to fully explore the potential differences within them. These "unknown class" signals may originate from different devices and exhibit distinct features, and directly classifying them as "unknown" prevents the system from making more refined identifications and processing of these signals. Therefore, further subdividing the unknown signals into multiple categories can help improve recognition accuracy and provide valuable references for future data annotation. This paper introduces a ss-k-means++ clustering algorithm based on cosine distance to further distinguish the identified unknown signals. The semi-supervised clustering algorithm leverages a small amount of labeled data, guiding the clustering process of unknown signals by incorporating known category information, thereby effectively distinguishing potential categories. It is also able to correct misclassified known classes, as unknown class data may contain misidentified known classes. The algorithm consists of two parts: estimating the optimal number of clusters and performing semi-supervised clustering. Most clustering algorithms require a predefined K value, which represents the number of clusters. However, this is not feasible in an open-world scenario, where an automatic method for estimating K is required. This paper employs the Brent algorithm to estimate the optimal number of clusters K for the ss-k-means++ clustering task. The Brent algorithm [45] is a univariate function optimization method that combines golden section search and parabolic interpolation, known for its speed and efficiency. In our method, the Brent algorithm is used to optimize the estimation of K, accelerating the search process. The ss-k-means++ algorithm [46] combines the advantages of semi-supervised learning [47] and k-means++ [48], allowing for clustering with partially labeled data. The core idea is to leverage the labeled data to guide the selection of initial centers, while clustering iteratively incorporates the unlabeled data. Let the labeled sample set be $E_a^t = \{(z_i^a, y_i^a)\}_{i=1}^{N_a}$, the unlabeled sample set be $Z_0^t = \{z_j^0\}_{j=1}^{M_0}$, and the total estimated number of classes be $\hat{C}_t$. First, for each known class $j \in C_l^t$, calculate the initial cluster center using the labeled data:

Algorithm 1 Few-shot incremental open-set recognition for OFSCIL

Require:
- Training dataset of the initial class D_b;
- Incremental dataset: D_n for each increment $n \in \{1, 2, \ldots, t\}$;
- Parameters of Feature extractor and Cosine similarity classifier θ, θ';

for $n = 0$ **to** t **do**
  if $n == 0$ **then**
    Step 1: Initial training
    Class Augmentation through Eq.9.
    Initialize θ and θ'.
    for each epoch $E = 1$ to ep **do**
      for each batch (x, y) in D_b **do**
        Forward propagation:
        $\hat{y} = \text{cos\_sim}(f(x; \theta), \theta')$.
        Compute loss $\mathcal{L}$ as Eq.8.
        Backward propagation:
        Update the parameters θ and θ'.
      end for
      Update prototype vectors by Eq.7.
    end for
    return Feature extractor ϕ_θ
  end if

  Step 2: Prototype generation and calibration
  if $n == 0$ **then**
    Generate initial class prototypes through Eq.12:
    $\phi(c_b) \leftarrow \phi(c_{b_i}), c_{b_i} \in D_b$
    Update the knowledge of classifier: $\mathcal{K} \leftarrow \phi(c_b)$
  else
    For each increment $n \in \{1, 2, \ldots, t\}$, perform:
    Generate incremental class prototypes for D_n through Eq.11:
    $\phi(c_n) \leftarrow \phi(c_{n_i}), c_{n_i} \in D_n$.
    Calibrate the incremental class prototype by Eq.18:
    $\phi(\bar{c}_n) \leftarrow \phi(c_n)$.
    Update the knowledge of classifier: $\mathcal{K} \leftarrow \mathcal{K} \cup \phi(\bar{c}_n)$
  end if

  Step 3: Open-set recognition and classification
  Open-set recognition according to Eq.24.
  Use prototype classifier for identification through Eq.19.
  return Prediction Result $\hat{y}$
end for

$$c_j = \frac{1}{|C_j|} \sum_{i: y_i^a = j} z_i^a, \tag{25}$$

where $|C_j|$ represents the sample count in the class j. For the remaining $\hat{C}_t - |C_l^t|$ new classes, the k-means++ algorithm is applied to select the initial centers from the unlabeled data Z_0^t. The selection probability for each sample z is given by:

$$P(z) = \frac{D(z)^2}{\sum_{z' \in Z_0^t} D(z')^2}, \tag{26}$$

This article has been accepted for publication in IEEE Transactions on Cognitive Communications and Networking. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TCCN.2025.3565589

where $D(z)$ represents the squared cosine distance between the sample z and the nearest selected center. After initializing the centers, each unlabeled sample is assigned to its nearest center, while labeled samples are assigned to their corresponding class centers. The center points are then iteratively updated by minimizing the total squared distances linking the data points with their assigned cluster centers, as shown in Eq.27. This process is repeated until convergence.

$$c_j = \frac{1}{|C_j|} \sum_{i \in C_j} z_i, \tag{27}$$

where C_j represents all the samples in the j-th cluster, including both labeled and unlabeled samples. The ss-k-means++ algorithm fully leverages the information from labeled data by incorporating the centers of known classes during initialization, which improves clustering accuracy. At the same time, it takes advantage of the k-means++ algorithm to effectively select initial centers from the unlabeled data, avoiding the instability caused by random initialization.

The class estimation algorithm uses the Brent algorithm to optimize the search for the number of clusters K, with the search range defined as $(|C_l^t|, |C_{max}^t|]$, where $|C_l^t|$ is the number of known classes, and $|C_{max}^t|$ is the expected maximum number of classes, which can be set to a large value. For each value of K during the search, the Average Clustering Accuracy (ACC) [49] and the Davies-Bouldin Score (DBS) [50] are calculated for the ss-k-means++ clustering, and their negative sum is returned. The process continues to optimize and find the optimal number of clusters. ACC is used to evaluate the clustering performance and is calculated as follows:

$$\text{ACC} = \max_{g \in G} \frac{1}{L} \sum_{i=1}^{L} 1\{y_i^v = g(y_i^{v'})\}, \tag{28}$$

where g is the set of all possible label assignments, y_i^v and $y_i^{v'}$ represent the true labels and the assigned cluster labels of each feature in the validation set, respectively. DBS is used to assess the quality of clustering by measuring the proportion of within-cluster distances to between-cluster distances, the calculation method is shown in follows:

$$\text{DBS} = \frac{1}{K} \sum_{i=1}^{K} \max_{j \neq i} \frac{S_i + S_j}{M_{ij}}, \tag{29}$$

where K represents the count of clusters obtained from clustering, S_i is the average intra-cluster distance for cluster i, and M_{ij} is the distance between clusters i and j. After estimating the optimal value of K, the ss-k-means++ model is used to cluster the data. The process for the unknown signal clustering algorithm is outlined in Algorithm 2.

Algorithm 2 Detected unknown signals cluster

Require: Few Labeled set of known class features: $E_a^t = \{(z_i^a, y_i^a)\}_{i=1}^{N_a}$;
Unlabeled set of unknown class features: $Z_0^t = \{z_i^0\}_{i=1}^{M_0}$;
Step 1: Estimating the Number of Classes
Optimize the number of clusters:

$$\hat{C}_t = \arg \min_{|C_l^t| \leq k \leq |C_{max}^t|} -(\text{ACC}(E_a^t, k) + \text{DBS}(Z_0^t, k)).$$

return *Best k* $\hat{C}_t$
Step 2: Semi-Supervised Clustering
Perform clustering according to Eq.25, Eq.26, Eq.27 :

$$\{y_i^0\} = \text{ss-k-means++}(E_a^t, Z_0^t, \hat{C}_t).$$

Group the clustered features, C is the set containing all cluster indices, $C = \{ 1, 2, \ldots, \hat{C}_t \}$:

$$Z_c^g = \{z_i^0 \mid y_i^0 = c\}, \quad \forall c \in C.$$

return $Z_i^g = \{Z_c^g \mid c \in C\}$

V. Experimental Results and Analysis

This paper uses the publicly available WiSig signal dataset [51], which contains 130 WiFi signal classes, with more than 50 samples per class. The dataset captures WiFi signals at a sampling rate of 25 Msps, with a center frequency of 2462 MHz and a bandwidth of 20 MHz. In the initial training phase, 80% of the samples make up the entire training set and 20% of the samples make up the test set. In the incremental phase, only 5 samples of each class are used for learning and the rest of the samples form the test set.

A. Evaluation metrics and experimental setup

We use the F1-score [52] to evaluate the performance of OSR. Normalized Mutual Information (NMI) [53] is utilized to assess the performance of clustering for unknown classes, and Harmonized Accuracy (HA) [44] is employed to assess the performance of FSCIL.

The F1-score, an essential metric for evaluating OSR performance, is calculated as the harmonic mean of Precision and Recall. The F1-score is calculated using the following formula:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \tag{30}$$

where Precision denotes the ratio of correctly identified positive samples among those predicted as positive, and Recall indicates the proportion of correctly predicted positive samples compared to the total actual positive cases. They are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{31}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{32}$$

where TP represents the count of samples for which the model correctly predicted a positive outcome, matching the true positive label. FP indicates the count of samples for which the model predicted a positive outcome, although the actual label was negative. FN Indicates the count of samples where the actual label is positive, but the model prediction is negative.

NMI is used to measure the similarity between two clusterings. It addresses the comparison problem of Mutual Information (MI) across different cluster sizes. Its formula is as follows:

$$\text{NMI}(U, V) = \frac{2 \times I(U; V)}{H(U) + H(V)}, \tag{33}$$

where U represents the clustering result from the algorithm, and V represents the ground truth clustering. $I(U;V)$ is the mutual information, which is computed as:

$$I(U;V) = \sum_{u\in U}\sum_{v\in V} P(u,v)\log\frac{P(u,v)}{P(u)P(v)}, \tag{34}$$

where the joint probability $P(u,v)$ represents the likelihood that a sample belongs to both cluster u and v, and $P(u)$ and $P(v)$ are the marginal probabilities of a sample belonging to clusters u and v, respectively. In Eq.33, $H(U)$ and $H(V)$ represent the entropy of the clustering results U and V, respectively.

In the task of FSCIL, due to the high proportion of base classes compared to incremental classes, a model that performs well on base classes but poorly on incremental classes can still have a good average accuracy. HA is a more accurate metric for evaluating the performance of FSCIL. Its formula is as follows:

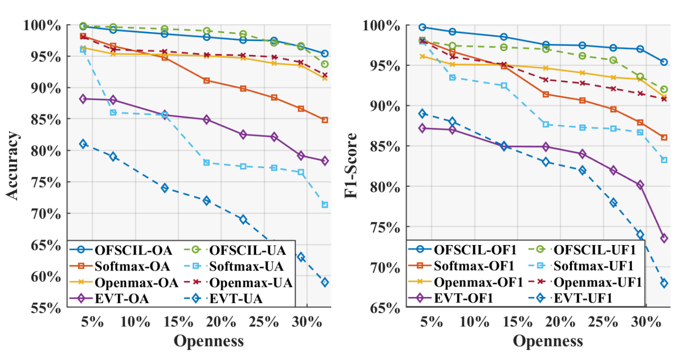
$$HA = \frac{2\times A_{\text{old}}\times A_{\text{new}}}{A_{\text{old}}+A_{\text{new}}}, \tag{35}$$

where A_{old} represents the average accuracy for the base classes, and A_{new} represents the average accuracy for the incremental classes. A balanced incremental classifier should perform well on both average accuracy and harmonic accuracy.

All experiments in this paper are implemented using Pytorch and trained on an NVIDIA GeForce GTX 3060. The batch size is set to 64, the learning rate to 0.001, and the optimizer used is adam. The number of epochs was initially set to 200, with early stopping applied.

B. Performance evaluation

1) Performance of OSR: In this section, we evaluate the performance of different methods in OSR scenarios with different levels of openness. The evaluation metrics include overall accuracy (OA), overall F1-score (OF1), Uknown class accuracy (UA) and unknown class F1-score (UF1). The detailed results are shown in Fig.5. We compare the OFSCIL proposed in this paper with the maximum softmax probability proposed in [17], an Extreme Value Theory (EVT) based countermeasure proposed in [18] and the Openmax classifier used in [54].



(a) (b)

Fig. 5: The performance of different methods on the OSR task. (a) Accuracy (b) F1-score

In the OSR experiments, 60 classes are considered known. As shown in the figure, OFSCIL maintains good performance across various evaluation metrics. Although the open-set recognition accuracy decreases as the degree of openness increases, the model still performs well. Specifically, when the openness is 22.5%, the overall accuracy reaches 97.36%. Even when the openness expands to more than 30%, where the quantity of unknown classes surpasses that of known training classes, the performance remains high with an overall accuracy of 95.38%. This demonstrates the high reliability and stability of OFSCIL, which adapts well to open-set scenarios. OFSCIL also showed strong competitiveness compared to other methods, with an overall accuracy of 3.95% higher than the next best method and an OF1 of 4.27% higher.

TABLE II: Clustering performance of different clustering algorithm, N indicates the number of unknown classes

N	5	10	20	30	40
HDBSCAN	57.2%	53.17%	50.8%	43.9%	40.8%
FINCH	62.9%	61.6%	58.1%	53.6%	51.7%
GCD	57.7%	56.8%	53.8%	52.2%	51.3%
OFSCIL(ours)	71.2%	64.7%	61.9%	61.8%	60.2%

Table II shows the clustering performance of different methods under different numbers of unknown emitter categories. It can be observed that the ss-k-means++ method based on cosine distance in OFSCIL performs better compared to other clustering methods: Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) [55], First Integer Neighbor Clustering Hierarchy (FINCH) [56] and Generalized Category Discovery (GCD) [49]. Compared to other methods, the NMI of our method improved by approximately 14% to 8.3%, demonstrating the ability to effectively distinguish unknown signal classes.

Table III shows the effectiveness of the proposed algorithm for evaluating different numbers of unknown categories. We found that OFSCIL can get close to the true number of unknown classes.

TABLE III: Estimation of the number of unknown classes

N	5	10	20	30	40
truth	65	70	80	90	100
estimate	67	72	81	93	114

2) Performance of FSCIL:

a) Comparison of FSCIL performance: In the experimental setup, the initial number of classes is 60, with 5 new classes added at each incremental stage. Each signal class contains only 5 samples, gradually increasing the total to 130 classes over 14 incremental steps. We compare the performance of our method against other class-incremental learning approaches utilized for signal recognition, namely FT [57], KD [28], CIL [23], ER [58], and CDIL [27]. FT is a typical fine-tuning method that freezes the feature extraction layer of the model and updates the classification layer with new data. KD stands for knowledge distillation, where the predictions from the old model serve as soft labels to steer the training process of the new model, preserving old knowledge. CIL combines knowledge distillation with EWC,

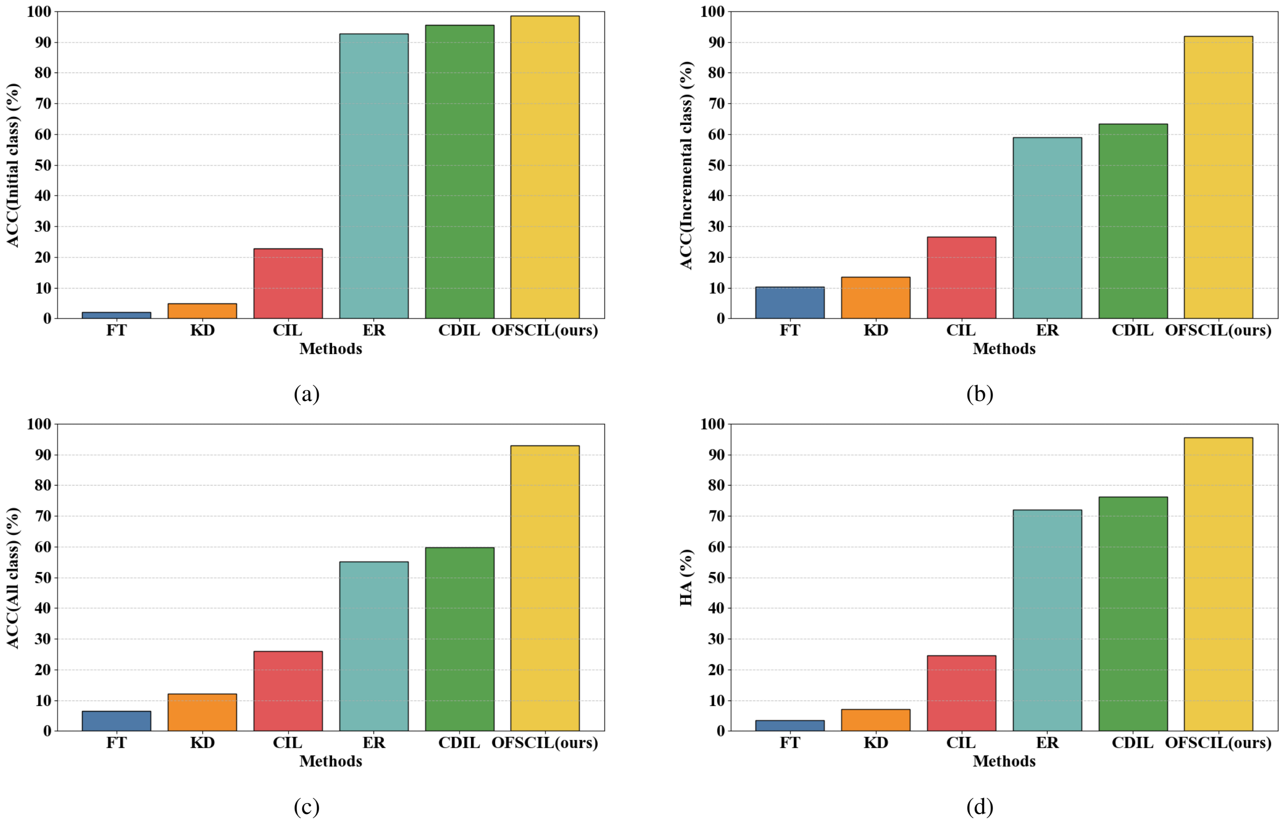


Fig. 6: Comparison of FSCIL performance across different CIL Methods in signal recognition

which protects the important knowledge of previous tasks by constraining changes to key parameters, thus mitigating catastrophic forgetting. ER mitigates forgetting of old classes by storing and replaying a portion of the old data. CDIL also replays some old data and designs a specialized distillation loss function for model updates. All methods were compared using the same dataset and experimental settings, with ER and CDIL replaying approximately 20% of old class samples. The comparison of FSCIL performance is shown in Fig.6. As shown in Fig.6, the existing incremental learning methods for signal recognition do not perform satisfactorily in FSCIL tasks. FT and KD exhibit very low performance, both for the initial classes and the newly added incremental classes, with overall accuracy and HA far below the performance requirements for FSCIL. In contrast, CIL improves performance by protecting important weights for old tasks during incremental learning, but it still falls short. ER and CDIL help retain knowledge of previously learned classes and demonstrate higher accuracy for the initial classes by replaying some past samples during incremental learning. Their accuracy for incremental classes is also much higher than that of FT and KD. ER and CDIL achieve overall accuracy of 55.23% and 59.74%, alongside HA of 72.13% and 76.21%, marking a notable improvement over FT and KD. The proposed OFSCIL is specifically designed for FSCIL tasks, and it outperforms all other methods in terms of incremental class accuracy, overall accuracy, and HA. Compared to the second ranked method, OFSCIL has improved incremental class accuracy, overall accuracy and HA by 28.71%, 33.37% and 18.82% respectively, making it the state-of-the-art in FSCIL for SEI.

b) Comparison of different initial class numbers: OFSCIL relies on a well-performing feature extractor to construct accurate feature prototypes. This section examines the effect of the number of initial training classes on the performance of subsequent FSCIL. We set the initial number of training classes to 40, 50, 60, 70, and 80, and increase 5 classes in each incremental step, with 5 samples per class, until reaching a total of 130 classes.

TABLE IV: FSCIL performance with different initial class numbers

Initial classes	40	50	60	70	80
New	90.57%	91.41%	92.06%	92.95%	93.83%
Overall	91.18%	92.29%	93.11%	94.27%	95.02%
HA	94.06%	94.41%	95.03%	95.41%	95.97%

Table IV shows the FSCIL performance with different initial class numbers. It can be observed that the number of initial classes has a certain impact on accuracy. Since DL is data-driven, having more initial training classes means a stronger feature extractor can be trained, leading to more accurate feature extraction, which is crucial for prototype construction.

c) FSCIL performance under normal samples conditions: This section explores the performance of OFSCIL under normal samples where there are sufficient sample of the incremental categories. As shown in the Table V. When sufficient samples are available, the feature extractor is better able to extract the features of new classes and generate more representative feature prototypes. OFSCIL has achieved promising results.

TABLE V: FSCIL performance under normal samples

Incremental classes	**30**	**40**	**50**	**60**	**70**
New(%)	95.33	95.14	95.00	94.80	94.67
Overall(%)	97.73	97.62	97.41	97.33	97.05
HA(%)	96.95	96.77	96.61	96.81	96.62

d) FSCIL performance under extreme sample conditions: This section explores the performance of OFSCIL under extreme conditions where there is only one sample of the incremental category. As shown in the Table VI. When there is only one sample, it is no longer possible to build a prototype for the new class. The newly added class is random in the feature space, which may deviate from its feature center, causing confusion among the central features of different classes.

TABLE VI: FSCIL performance under the one-shot condition

Incremental classes	**30**	**40**	**50**	**60**	**70**
New(%)	81.38	81.33	80.70	80.17	79.81
Overall(%)	86.72	85.02	84.79	83.49	83.15
HA(%)	89.23	89.16	88.57	88.43	88.04

3) Performance of FIOSR:

a) Performance analysis: This section primarily evaluates the effectiveness of OFSCIL in handling FIOSR task, which require the system to continually learn new classes while retaining the ability to recognize and exclude unknown classes that have not been previously encountered or trained on. The experiment begins with 60 initial classes, with incremental class numbers set to 5, 10, 20, 30, 40 and 50 classes, with 5 samples per class. The remaining classes are treated as unknown. The test set for the experiment includes not only the initially learned classes and the newly added incremental classes but also the remaining unknown classes. Fig.7 shows the performance of OFSCIL under the FIOSR task. KA in the figure indicates the Known class accuracy.

As the number of incremental classes increases, the open-set pressure on the model gradually decreases due to the corresponding reduction in the number of unknown classes. At the same time, the introduction of incremental classes presents a greater learning challenge for the model, as it must not only acquire new classes but also retain knowledge of previously learned classes. The experimental results show that OFSCIL exhibits some fluctuations in performance when handling FIOSR task. In the scenario with 5 incremental classes, the OF1 reaches 95.31%, with 65 unknown classes remaining which account for 50% of the total classes. As more new classes are added, the model's performance gradually declines to 88.12%. The experimental results underscore the persistent challenge of alleviating catastrophic forgetting in incremental learning. The introduction of new classes increases feature space complexity, leading to competition between features of old and new classes and a gradual decline in classification performance. While the reduction in unknown classes lessens open-set pressure , the proximity or overlap of prototypes from new and previously learned classes complicates inter-class distance measurements, intensifying the difficulty of open-set classification. Overall, OFSCIL exhibits strong robustness, effectively addressing the dual challenges of assimilating new classes and managing open-set pressure in complex incremental learning environments.

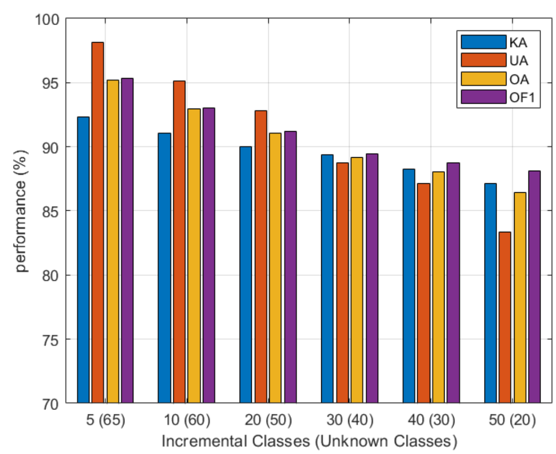


Fig. 7: The performance of OFSCIL under the FIOSR task

b) Effectiveness of different modules: The proposed OFSCIL framework includes a decoupling strategy and designs a prototype classifier as an alternative, an integrated loss function combining angular penalty loss and prototype loss, class augmentation and prototype calibration. Table VII is listed the effectiveness of each module on the performance of OFSCIL under the FSCIL and FIOSR task. For FSCIL, the initial class is set to class 60 and increases by 5 classes each time until it increases to class 130. For FIOSR, the initial class is also set to 60 classes, and the incremental classes are gradually increased to 50.

We can observe that for FSCIL, when the decoupling strategy is not used and the prototype classifier is unemployed, the model is unable to acquire new classes, and catastrophic forgetting occurs. In contrast, the feature extractor is trained with enhanced angular penalty loss, and the prototype classifier

TABLE VII: Effectiveness of different modules

Component				FSCIL			FIOSR	
Prototype classifier	**Integrated loss**	**Class augmentation**	**Prototype calibration**	**New**	**Overall**	**HA**	**OA**	**OF1**
×	×	×	×	0.72%	45.85%	1.42%	56.62%	58.85%
✓	×	×	×	87.92%	89.16%	91.33%	80.23%	81.05%
✓	✓	×	×	89.69%	91.15%	93.42%	82.21%	84.56%
✓	✓	✓	×	91.30%	92.45%	94.43%	85.11%	86.75%
✓	✓	✓	✓	92.06%	93.11%	95.03%	86.44%	88.12%

is used to learn new class features effectively, and the frozen feature extractor alleviates catastrophic forgetting. Adding prototype losses further improves accuracy. Specifically, for the 14 incremental steps, the accuracy for the incremental classes is 89.69%, the OA is 91.15%, and the HA is 93.42%. Compared to using only the enhanced angular penalty loss, these values show improvements of 1.77%, 1.99%, and 2.09%, respectively. Class augmentation further enhances the performance of FSCIL. Specifically, the accuracy for the incremental classes increases by an additional 1.61%, the OA improves by 1.3%, and HA increases by 1.01%. Finally, after prototype calibration, OFSCIL achieved incremental accuracy, OA and HA of 92.06%, 93.11% and 95.03%, respectively.

For FIOSR, without using a decoupling strategy and open-set classifier, it is almost impossible to identify new classes. The use of open-set classifier which is inherited from the prototype classifier is effective in identifying new classes and discovering unknown classes. The integration loss, class augmentation and prototype calibration further improved OF1 by 7.07%.

C. Complexity analysis

The complexity of OFSCIL mainly arises from the carefully designed feature extraction network used during the initial training phase. Table VIII lists the parameters and floating-point operations (FLOPs) of the feature extractor, which are 0.024M and 2.1M, respectively. Additionally, the table VIII shows the inference times on both GPU and CPU. It can be observed that, due to the low computational complexity, the proposed method is suitable for deployment on practical devices.

TABLE VIII: Feature extractor complexity analysis

Params(M)	**FLOPS(M)**	**Infer/GPU(ms)**	**Infer/CPU(ms)**
0.024	2.1	0.43	1.03

Table IX demonstrates the execution time of different incremental learning methods. For FSCIL, unlike other methods that require retraining with new data to update the entire model's parameters (such as KD, ER, CIL, and CDIL) or updating the classifier with new data (as in FT), OFSCIL freezes the feature extractor and uses the prototype classifier. This corresponds to the inference phase of the model. The algorithmic complexity is much less than the methods that have to update the model with incremental classes of data. The reduction in execution time of OFSCIL relative to other methods is about 75%, 78%, 87%, 91% and 97%, respectively.

TABLE IX: Method complexity comparison

Method	**FT**	**KD**	**CIL**	**ER**	**CDIL**	**OFSCIL (ours)**
Execution time(s)	10.65	12.32	21.56	32.10	87.53	2.62

Table X presents the performance achieved by the proposed OFSCIL framework across its three stages along with the required time. The number of training epochs in the initial training stage is 200; the incremental learning stage shows the execution time required for 30 incremental classes, which corresponds to 6 incremental steps; the final recognition stage displays the inference time required for a single sample, under the conditions of 40 unknown classes. The results in Table X show that OFSCIL can achieve a good accuracy rate with low computational consumption, achieving a balance between computational efficiency and accuracy.

TABLE X: Method complexity comparison

Stage	Performance	Efficiency
Initial training	99%	235.51s
Incremental Learning	94.41%	1.36s
Identify	88.87%	0.43ms

VI. Conclusion

This paper provides a comprehensive analysis of the challenges faced by SEI systems in a dynamic open-world and outlines three tasks that SEI systems need to address: OSR, FSCIL and FIOSR. To address the challenge, this paper proposes a framework called OFSCIL as a solution for SEI, aiming to provide a more robust paradigm for SEI in dynamic open environments. OFSCIL decouples the model's feature extractor from the classifier, focusing on training a strong feature extractor. To tackle the OSR challenge, this paper designs a prototype distance-based open-set classifier for effectively identifying unknown emitters. Additionally, a ss-k-means++ algorithm based on cosine distance is introduced to further distinguish unknown signals. For the FSCIL challenge, we propose an innovative combinatorial loss function combined with a class enhancement approach to train the feature extractor and mitigate the catastrophic forgetting problem by freezing feature extractors and designing a sustainably evolving prototype classifier. A prototype calibration method is proposed to alleviate the overfitting problem in few-shot scenarios. Ultimately, OFSCIL effectively adapts to OSR, FSCIL and FIOSR task. Extensive experiments on Wisig datasets and complexity analysis demonstrate its capability to balance computational efficiency and accuracy. In the future, we aim to explore more advanced open-set classifiers, clustering algorithms, and prototype calibration strategies to further drive SEI systems toward a more dynamic open-set world.

References

[1] J. Shi, B. Ge, H. Jiang, R. Yang, G. Si, Y. Wang, G. Gui, and Y. Lin, "Feature-imitation federated learning: An efficient approach for specific emitter identification in low-resource environments," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 6, pp. 2061–2075, Dec. 2024.

[2] P. Tang, G. Ding, Y. Xu, Y. Jiao, Y. Song, and G. Wei, "Causal learning for robust specific emitter identification over unknown channel statistics," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 5316–5329, Apr. 2024.

[3] L. Xu, X. Fu, Y. Wang, Q. Zhang, H. Zhao, Y. Lin, and G. Gui, "Enhanced few-shot specific emitter identification via phase shift prediction and decoupling," *IEEE Transactions on Cognitive Communications and Networking*, Jul. 2024.

[4] J. Zhang, Y. Liu, G. Ding, B. Tang, and Y. Chen, "Adaptive decomposition and extraction network of individual fingerprint features for specific emitter identification," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 8515–8528, Jul. 2024.

[5] A. Bendale and T. Boult, "Towards open world recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 1893–1902.

[6] Y. Peng, P. Liu, Q. Zhang, L. Guo, Y. Liu, Y. Wang, Y. Lin, and G. Gui, "Dynamic adaptation rff identification method leveraging cognitive representation learning," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 7939–7951, Aug. 2024.

[7] X. Zhang, Y. Huang, M. Lin, Y. Tian, and J. An, "Transmitter identification with contrastive learning in incremental open-set recognition," *IEEE Internet of Things Journal*, vol. 11, no. 3, pp. 4693–4711, Feb. 2024.

[8] W. Liu, X. Nie, B. Zhang, and X. Sun, "Incremental learning with open-set recognition for remote sensing image scene classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–16, May 2022.

[9] X. Ma, K. Ji, S. Feng, L. Zhang, B. Xiong, and G. Kuang, "Open set recognition with incremental learning for sar target classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–14, Jun. 2023.

[10] N. Yang, B. Zhang, G. Ding, Y. Wei, G. Wei, J. Wang, and D. Guo, "Specific emitter identification with limited samples: A model-agnostic meta-learning approach," *IEEE Communications Letters*, vol. 26, no. 2, pp. 345–349, Sep. 2022.

[11] S. Tian, L. Li, W. Li, H. Ran, X. Ning, and P. Tiwari, "A survey on few-shot class-incremental learning," *Neural Networks*, vol. 169, pp. 307–324, Jan. 2024.

[12] M. Zhang, P. Tang, G. Wei, X. Ni, G. Ding, and H. Wang, "Open set domain adaptation for automatic modulation classification in dynamic communication environments," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 3, pp. 852–865, Mar. 2024.

[13] C. Geng, S. Huang, and S. Chen, "Recent advances in open set recognition: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 10, pp. 3614–3631, Mar. 2021.

[14] S. Huang, L. Guo, X. Fu, Y. Peng, Y. Guo, Y. Wang, Q. Zhang, G. Gui, and H. Sari, "Open-set specific emitter identification leveraging enhanced metric denoising auto-encoders," *IEEE Internet of Things Journal*, vol. 12, no. 4, pp. 3453–3462, May 2024.

[15] L. Guo, C. Liu, Y. Liu, Y. Lin, and G. Gui, "Towards open-set specific emitter identification using auxiliary classifier generative adversarial network and openmax," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 6, pp. 2019–2028, Jun. 2024.

[16] X. Zhang, M. Lin, Y. Tian, Y. Huang, J. An, and T. Cui, "Data-enhancement-aided protocol-agnostic transmitter recognition for open-set in iot," *IEEE Internet of Things Journal*, vol. 10, no. 10, pp. 8630–8644, May 2023.

[17] S. Hanna, S. Karunaratne, and D. Cabric, "Deep learning approaches for open set wireless transmitter authorization," in *IEEE 21st International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, 2020, pp. 1–5.

[18] X. Zhang, T. Li, P. Gong, R. Liu, X. Zha, and W. Tang, "Open set recognition of communication signal modulation based on deep learning," *IEEE Communications Letters*, vol. 26, no. 7, pp. 1588–1592, May 2022.

[19] G. M. Van de Ven, T. Tuytelaars, and A. S. Tolias, "Three types of incremental learning," *Nature Machine Intelligence*, vol. 4, no. 12, pp. 1185–1197, Dec. 2022.

[20] X. Shen, T. Zhang, H. Wu, X. Qiao, Y. Du, and G. Gui, "An adaptive domain-incremental framework with knowledge replay and domain alignment for specific emitter identification," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 3519–3533, Mar. 2025.

[21] Z. Fan, Y. Tu, Y. Lin, and Q. Shi, "Class-incremental learning for recognition of complex-valued signals," *IEEE Transactions on Cognitive Communications and Networking*, vol. 10, no. 2, pp. 417–428, Apr. 2024.

[22] X. Xu, W. Wang, and J. Wang, "A three-way incremental-learning algorithm for radar emitter identification," *Frontiers of Computer Science*, vol. 10, pp. 673–688, Dec. 2016.

[23] Y. Liu, J. Wang, J. Li, S. Niu, and H. Song, "Class-incremental learning for wireless device identification in iot," *IEEE Internet of Things Journal*, vol. 8, no. 23, pp. 17 227–17 235, Dec. 2021.

[24] M. Phuong and C. Lampert, "Towards understanding knowledge distillation," in *International conference on machine learning (ICML)*, 2019, pp. 5142–5151.

[25] J. Kirkpatrick, R. Pascanu, N. Rabinowitz, J. Veness, G. Desjardins, A. A. Rusu, K. Milan, J. Quan, T. Ramalho, A. Grabska-Barwinska, D. Hassabis, C. Clopath, D. Kumaran, and R. Hadsell, "Overcoming catastrophic forgetting in neural networks," *Proceedings of the National Academy of Sciences*, vol. 114, no. 13, pp. 3521–3526, 2017.

[26] A. Chaudhry, P. K. Dokania, T. Ajanthan, and P. H. Torr, "Riemannian walk for incremental learning: Understanding forgetting and intransigence," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 532–547.

[27] M. Liu, J. Wang, N. Zhao, Y. Chen, H. Song, and F. R. Yu, "Radio frequency fingerprint collaborative intelligent identification using incremental learning," *IEEE Transactions on Network Science and Engineering*, vol. 9, no. 5, pp. 3222–3233, Oct. 2022.

[28] J. Zhou, Y. Peng, G. Gui, Y. Lin, B. Adebisi, H. Gacanin, and H. Sari, "A novel radio frequency fingerprint identification method using incremental learning," in *IEEE 96th Vehicular Technology Conference (VTC-Fall)*, 2022, pp. 1–5.

[29] F. Shi, H. Wan, Z. Feng, X. Fu, Q. Wang, Q. Xuan, Y. Lin, and G. Gui, "Enhanced radio frequency fingerprint identification using length-robust representation and incremental learning," *IEEE Internet of Things Journal*, 2025.

[30] D. Li, J. Qi, S. Hong, P. Deng, and H. Sun, "A class-incremental approach with self-training and prototype augmentation for specific emitter identification," *IEEE Transactions on Information Forensics and Security*, vol. 19, pp. 1714–1727, 2024.

[31] X. Han, S. Chen, M. Xie, and J. Yang, "Prototype-based method for incremental radar emitter identification," *IET Radar, Sonar & Navigation*, vol. 17, no. 7, pp. 1105–1114, Apr. 2023.

[32] T. Zhao, S. Sarkar, Y. Tian, and D. Cabric, "Anomaly transmitter recognition and tracking," in *2024 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, 2024, pp. 357–364.

[33] D. Li, X. Chen, S. Hong, J. Qi, and H. Sun, "Fscil-sei: Few-shot class-incremental learning approach for specific emitter identification," *IEEE Transactions on Instrumentation and Measurement*, vol. 74, pp. 1–14, 2025.

[34] T. Li, Z. Wen, Y. Jiang, J. Wang, J. Su, Z. Hong, L. Yu, and S. He, "Meta-rff: Few-shot open-set incremental learning for rf fingerprint recognition via multi-phase meta task adaptation," in *Wireless Artificial Intelligent Computing Systems and Applications(WASA)*, 2025, pp. 221–234.

[35] W. Liu, Z. Liu, Z. Yu, B. Dai, R. Lin, Y. Wang, J. M. Rehg, and L. Song, "Decoupled networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition(CVPR)*, 2018, pp. 2771–2779.

[36] Q. Wang, D. Zhou, Y. Zhang, D. Zhan, and H. Ye, "Few-shot class-incremental learning via training-free prototype calibration," in *Advances in Neural Information Processing Systems(NIPS)*, vol. 36, 2023, pp. 15 060–15 076.

[37] W. Liu, Y. Wen, Z. Yu, M. Li, B. Raj, and L. Song, "Sphereface: Deep hypersphere embedding for face recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*, 2017, pp. 6738–6746.

[38] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "Arcface: Additive angular margin loss for deep face recognition," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition(CVPR)*, 2019, pp. 4685–4694.

[39] J. Snell, K. Swersky, and R. Zemel, "Prototypical networks for few-shot learning," in *Advances in neural information processing systems (NIPS)*, vol. 30, 2017.

[40] H. Yang, X. Zhang, F. Yin, and C. Liu, "Robust classification with convolutional prototype learning," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 3474–3482.

[41] F. Zhou, P. Wang, L. Zhang, W. Wei, and Y. Zhang, "Revisiting prototypical network for cross domain few-shot learning," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition(CVPR)*, 2023, pp. 20 061–20 070.

[42] H. Zhang, M. Cisse, Y. N. Dauphin, and D. LopezPaz, "mixup: Beyond empirical risk minimization," in *International Conference on Learning Representations(ICLR)*, 2018.

[43] F. Zhu, Z. Cheng, X. Zhang, and C. Liu, "Class-incremental learning via dual augmentation," in *Advances in Neural Information Processing Systems (NIPS)*, vol. 34, 2021, pp. 14 306–14 318.

[44] C. Peng, K. Zhao, T. Wang, M. Li, and B. C. Lovell, "Few-shot class-incremental learning from an open-set perspective," in *European Conference on Computer Vision (ECCV)*. Springer, 2022, pp. 382–397.

[45] K. Han, S.-A. Rebuffi, S. Ehrhardt, A. Vedaldi, and A. Zisserman, "Autonovel: Automatically discovering and learning novel visual categories," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 10, pp. 6767–6781, 2022.

[46] A. Vouros and E. Vasilaki, "A semi-supervised sparse k-means algorithm," *Pattern Recognition Letters*, vol. 142, pp. 65–71, Feb. 2021.

[47] D. P. Kingma, S. Mohamed, D. Jimenez Rezende, and M. Welling, "Semi-supervised learning with deep generative models," in *Advances in neural information processing systems (NIPS)*, vol. 27, 2014.

[48] D. Arthur and S. Vassilvitskii, "k-means++: the advantages of careful seeding," in *Proceedings of the Eighteenth Annual ACM-SIAM Symposium on Discrete Algorithms*, 2007, pp. 1027–1035.

[49] S. Vaze, K. Hant, A. Vedaldi, and A. Zisserman, "Generalized category discovery," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 7482–7491.

[50] Y. A. Wijaya, D. A. Kurniady, E. Setyanto, W. S. Tarihoran, D. Rusmana, and R. Rahim, "Davies bouldin index algorithm for optimizing clustering case studies mapping school facilities," *TEM J*, vol. 10, no. 3, pp. 1099–1103, 2021.

[51] S. Hanna, S. Karunaratne, and D. Cabric, "Wisig: A large-scale wifi signal dataset for receiver and channel agnostic rf fingerprinting," *IEEE Access*, vol. 10, pp. 22 808–22 818, Feb. 2022.

[52] W. J. Scheirer, A. de Rezende Rocha, A. Sapkota, and T. E. Boult, "Toward open set recognition," *IEEE Transactions on pattern analysis and machine intelligence*, vol. 35, no. 7, pp. 1757–1772, Nov. 2012.

[53] A. Strehl and J. Ghosh, "Cluster ensembles—a knowledge reuse framework for combining multiple partitions," *The Journal of Machine Learning Research*, vol. 3, pp. 583–617, Mar. 2003.

[54] Y. Zhao, X. Wang, Z. Lin, and Z. Huang, "Multi-classifier fusion for open-set specific emitter identification," *Remote Sensing*, vol. 14, no. 9, 2022.

[55] L. McInnes and J. Healy, "Accelerated hierarchical density based clustering," in *IEEE International Conference on Data Mining Workshops (ICDMW)*, 2017, pp. 33–42.

[56] S. Sarfraz, V. Sharma, and R. Stiefelhagen, "Efficient parameter-free clustering using first neighbor relations," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR)*, 2019, pp. 8926–8935.

[57] N. Tajbakhsh, J. Y. Shin, S. R. Gurudu, R. T. Hurst, C. B. Kendall, M. B. Gotway, and J. Liang, "Convolutional neural networks for medical image analysis: Full training or fine tuning?" *IEEE Transactions on Medical Imaging*, vol. 35, no. 5, pp. 1299–1312, May 2016.

[58] Y. Zhao, S. Chen, T. Chen, W. Shen, S. Zheng, Z. Zhao, and X. Yang, "Incremental learning of radio modulation classification based on sample recall," *China Communications*, vol. 20, no. 7, pp. 258–272, Jul. 2023.

Tao Zhang received the B.S. degree in communication engineering and the Ph.D. degree in communication and information system from the College of Communications Engineering, PLA University of Science and Technology, Nanjing, China, in 2011 and 2016, respectively. He is currently an Associate Researcher with the Sixty-Third Research Institute, National University of Defense Technology. He has authored or coauthored several papers in IEEE Transactions on Communications, IEEE Transactions On Vehicular Technology, IEEE Transactions on Cognitive Communications and Networking, IEEE Internet of Things Journal. His current research interests include cooperative communications, wireless sensor networks, physical layer security, and cognitive radio systems.

Xiaoyu Shen is currently pursuing the M.S. degree with the Sixty-third Research Institute of the National University of Defence Technology and Nanjing University of Information Science and Technology, Nanjing, China. His current research interests include communication specific emitter identification, incremental learning, physical layer security, and deep learning.

Zhihui Shang received his B.S. and M.S. degrees from the Zhengzhou University of Light Industry, Zhengzhou, China, in 2014 and 2017, respectively, and the Ph.D. degree from the Army Engineering University of PLA, Nanjing, China, in 2021. He is currently a Postdoctoral Researcher with the Sixty-Third Research Institute, National University of Defense Technology, Nanjing, China, and he is also a teacher with the College of Medical Information Engineering, Zunyi Medical University, Zunyi, China. His research interests include spectrum management, spectrum prediction, cooperative communications, and cognitive radio systems. He serves as a Reviewer for IEEE Transactions on Wireless Communications and IEEE Transactions on Cognitive Communications and Networking.

Guoru Ding is currently a Professor with the College of Communications Engineering, Nanjing, China. He received the B.S. (Hons.) degree in electrical engineering from Xidian University, Xi'an, China, in 2008, and the Ph.D. (Hons.) degree in communications and information systems from the College of Communications Engineering, Nanjing, China, in 2014. From 2015 to 2018, he was a Post-Doctoral Research Associate with the National Mobile Communications Research Laboratory, Southeast University, Nanjing, China. His research interests include intelligent wireless communications and signal processing.

He is a recipient of a project supported by the National Science Fund for Distinguished Young Scholars of China. He has received the 14th IEEE COMSOC Aisa-Pacific Outstanding Young Researcher Award in 2019. He has served as a Guest Editor for the IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS and an Associate Editor of the IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING. He is now an Editor of CHINA COMMUNICATIONS.

Hao Wu received the M.S. degree in communication engineering from the College of Communications Engineering, PLA University of Science and Technology, Nanjing, China, in 2008, and the Ph.D. degree in communications and information systems with the College of Communications Engineering, Army Engineering University of PLA, Nanjing, in 2017. He is currently an Associate Professor with the Sixty-third Research Institute, National University of Defense Technology. His current research interest includes spectrum sensing, spectrum management, and communication anti-jamming.

Xiaoqiang Qiao received the B.S. and M.S. degrees in communications engineering and information system from the College of Communications Engineering, PLA University of Science and Technology, Nanjing, China, in 2002 and 2005, respectively. He is currently a Senior Engineer with the Sixty-Third Research Institute, National University of Defense Technology. His research interests include electromagnetic spectrum management and communication anti-jamming.